

When to Escalate: A Cost-Aware Belief Policy for Conversational Agents Under Hidden Intent

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Abstract

A conversational agent replying to inbound sales leads must choose an action before it knows the lead’s intent. It can answer, ask a qualifying question, hold, or escalate to a human, but the true buying-readiness and whether a human is actually needed are hidden. The two ways of being wrong are not equally costly. A missed escalation, where a lead who needed a person did not get one, is far more expensive than a needless one, and some wrong answers are not a cost to trade off at all but a line the agent must not cross. We build a small agent that makes this trade-off explicit. It holds a belief over the hidden state, split into readiness (hot, warm, cold) and a separate probability that the case needs a human, and at each message it picks the action with the lowest expected cost under a cost matrix that encodes the asymmetry. Escalation is split into notify, where a human is told while the conversation continues, and pause, where the agent stops, because the two carry different costs. The hardest wrong answers, such as false legal or land claims, are enforced as a hard constraint rather than a priced term.

We test on 100 synthetic cases with cached language-model beliefs, comparing the cost-aware policy against the same policy with a uniform cost matrix and against three fixed-action baselines. The cost-aware policy reaches a mean cost of 1.72 against 2.58 for the uniform-cost version, so the asymmetry does most of the work — though the uniform baseline turns out to select the same action as a plain 0.5 threshold on all 100 cases, and reweighting the case set toward the design’s own readiness prior moves the gap from 0.86 to 1.07 rather than closing it, so the result’s direction is robust to reweighting even as its magnitude shifts. Against a policy that escalates every message the result is close to a tie on cost, 1.72 against 1.74, but the cost-aware policy reaches it while escalating 43 times out of 100 rather than all 100, so the gain there is in human load, not cost. Every missed escalation traces to one cause: the be-

lief under-estimated the needs-human probability rather than misreading readiness. Expected calibration error on that marginal is 0.142 (bootstrap 95% CI [0.100, 0.249]), concentrated in the bins adjacent to the escalation threshold rather than the bin containing it; recalibrating the marginal in-sample halves the misses, from 16 to 8, and the residual 8 are traceable to the elicited probability taking only one-decimal values. We also report a failure the test set cannot produce by construction, a high-cost message dropped when it arrived batched with a routine one, observed in a live run. Code and data are available.

Problem statement. The agent observes an inbound message from a sales lead. It must select one action from {answer, ask a qualifying question, hold, escalate to a human} because the lead’s true intent and buying-readiness are not known.

Note on the action set. The statement above is the original problem seed, reproduced verbatim. In the course of this work escalation was split into two distinct actions: *escalate_notify*, where a human is alerted while the conversation continues, and *escalate_pause*, where the agent stops and hands over. The two are separated because they carry materially different costs, and collapsing them would hide that difference from the decision rule. The resulting five-action set — {answer, ask, hold, escalate_notify, escalate_pause} — is the one used throughout this paper and in every reported result.

1 Introduction

A conversational agent that answers inbound sales leads has to act before it knows who it is talking to. The same short message, “can I get more info”, might come from a serious buyer, a competitor fishing for pricing, or an automated blast. The agent sees the text. It does not see the intent behind it. It still has to do something.

The two ways of getting this wrong are not equally expensive, and that is the whole problem. An agent that escalates every uncertain message to a human never misses a real buyer, but it spends human attention on messages that never needed it. An agent that answers everything itself keeps humans free, but eventually gives a confident wrong reply to a

lead who needed a person, or worse, answers a question it had no business answering, like a request for legal or land documents. A naive confidence threshold treats these two failures as if they sat on the same scale. They do not. A missed hand-off to a human is far more costly than a needless one, and some wrong answers are not a cost to trade off at all but a line that must not be crossed.

This paper builds a small agent that makes the trade-off explicit. It maintains a belief over the lead’s hidden state, split into readiness (hot, warm, cold) and a separate probability that the case needs a human, and at each message it chooses the action with the lowest expected cost under a cost matrix that encodes the asymmetry directly. The action set has five members: answer, ask a qualifying question, hold, escalate-notify, and escalate-pause. Escalation is split into notify and pause because the two carry different costs. A needless pause can let a live lead go cold, while a needless notify only spends a human’s glance. The hardest wrong answers, false legal or land claims, are enforced not as a high price but as a hard constraint that removes the answer action entirely.

The contribution is not a new algorithm. The decision rule is a textbook one-step Bayes decision. The contribution is a careful and honest account of what cost-awareness actually buys on this problem, and where it does not help. Over 100 synthetic cases with cached language-model beliefs, the cost-aware policy reaches a mean cost of 1.72 against 2.58 for the same policy stripped of its cost asymmetry, a real gain from making the trade-off explicit. But against a policy that simply escalates everything, the cost is close to a tie, 1.72 against 1.74. What the cost-aware policy buys there is not lower cost but far less human load, reaching the same cost while escalating 43 times instead of 100. We report both comparisons rather than only the favourable one.

The failure analysis is where the design earns or loses trust. Every missed escalation traces to a single cause: the belief under-estimated the needs-human probability, and did not misread readiness. The model is measurably under-confident on that one marginal in two of the bins nearest the threshold, with an expected calibration error of 0.142 on needs-human, under-confident in exactly the range that suppresses escalation. This means the fix is recalibration, not a redesign of the state. We also report a failure the test set cannot even exhibit by construction, a high-cost message dropped when batched with a routine one at a turn boundary, observed in a live run and named as a limitation rather than hidden.

The rest of the paper is organized as follows. Section 2 places the work against learning-to-defer and selective-prediction literature and records the design changes that came from public discussion with practitioners. Section 3 gives the agent design. Section 4 gives the probability model and decision rule. Section 5 describes the test. Section 6 reports results, Section 7 the failure analysis, and Section 8 the limitations, ethics, and human-control considerations.

2 Related work and user discussions

2.1 Related work

The escalate action places this work in the *learning-to-defer* literature, where an agent chooses between deciding itself and

handing off to a human who has their own error and cost. Mozannar and Sontag [2020] give a formal, consistent treatment of this, and Madras *et al.* [2018] frame deferral with an eye to fairness and to the point that a human is not automatically the safer choice. This paper uses that framing but does not train a joint defer-classifier. The escalate action is priced in a cost matrix rather than learned, and the human is treated as a costed fallback rather than modelled with a per-case error rate.

The hold and ask actions are reject options in the sense of selective prediction, where a model acts on part of its input and abstains on the rest, trading coverage against risk [Geifman and El-Yaniv, 2017]. The policy here reports how often it hands off rather than forcing a decision, which is the coverage side of that trade.

The belief comes from a language model, so whether its probabilities can be thresholded at all depends on calibration. Guo *et al.* [2017] show that modern networks trained with a proper scoring rule are still often overconfident, and give the expected calibration error and reliability diagram used in Section 6. Overconfidence matters here in a specific way: it makes automated actions look cheaper than they are and suppresses escalation, which is exactly the failure the results show.

Finally, the problem is framed as a POMDP, where the true state is hidden and the agent acts on a belief [Kaelbling *et al.*, 1998]. Exact planning over belief space is intractable, so this work solves the one-step problem myopically: it minimises immediate expected cost under the current belief and does not plan over future beliefs. This is a deliberate scope choice, not an oversight, and it has a known cost of its own, discussed in Section 8: a strict one-step rule undervalues the ask action, whose payoff is a better belief on the next turn rather than an immediate gain.

2.2 User discussions

Several design choices came from posting the problem to practitioners rather than from the literature. The record of these exchanges is in the project’s discussion log; two of them changed the design in ways that the results later bore out.

A practitioner building message agents pointed out that treating an empty text field as the trigger for escalation is fragile, because reactions, stickers, captionless media, and system events all arrive as “a message with no text” and hit the same branch. The suggested fix was to classify the event type first and choose the action from the type, rather than inferring it from the absence of text. This maps directly onto a failure the paper reports in Section 7: a callback request that arrived batched with a routine message was dropped because the two were handled as one turn. The design response, a per-message scan for critical triggers before the turn is acted on, is the same idea in a different guise: distinct signals should not be collapsed into one decision.

A second exchange, on calibration, sharpened how the belief should be checked. The point raised was that training on a proper scoring rule pushes toward calibration but does not guarantee it, so calibration has to be measured rather than assumed, and measured where the decision actually turns rather than uniformly. This is what Section 6 does: the reported cal-

ibration error is on the needs-human marginal, in the probability range where the escalation threshold sits, because that is the range that decides whether a handoff happens.

Other exchanges shaped earlier design decisions without tying as directly to a reported result: separating “cannot parse the input” from “understood but not confident enough to help” as distinct reasons for escalation, and tagging each hold or escalation with its reason so the reasons can be priced differently. These are recorded in the discussion log with the change each one produced.

3 Agent design

The agent is defined by seven parts: what it observes, what it cannot see, the belief it holds over the hidden part, the actions available to it, the cost of a wrong action, the rule that picks an action, and the feedback it gets afterward.

Input. On each turn the agent receives the inbound message and the prior turns of the same conversation. It also receives a small amount of position context: the turn index and how many times a near-identical message has repeated. Roughly a third of the message types the agent must handle are position-dependent, so the same text can warrant a different action depending on where in the conversation it lands.

Hidden state. What the agent cannot see is split into two independent parts. The first is *readiness*, one of hot, warm, or cold, which is mutually exclusive: at any moment the lead really is in one of these. The second is whether the case *needs a human*, which is a separate property of the moment, not a fourth readiness level. The two are kept independent on purpose. If needs-human competed with hot, warm, and cold for the same probability mass, then evidence that a lead is hot would mechanically lower the estimated chance it needs a human, which is backwards: a hot, ready-to-buy lead is often exactly the one a person should handle. Section 6 reports that the failures are consistent with this independence rather than against it.

Belief. The belief is what the agent holds over the hidden state given everything seen so far: a distribution over hot, warm, cold that sums to one, and a separate probability that the case needs a human. The belief comes from a language model rather than a trained classifier, which makes calibration the load-bearing assumption of the whole design. A language model’s stated probabilities are often overconfident, and overconfidence makes automated actions look cheaper than they are, which suppresses escalation. So the belief is checked against labeled cases before any threshold is trusted, and the check is reported rather than assumed.

Actions. The agent chooses one of five: answer, ask a qualifying question, hold, escalate-notify (tell a human while the conversation continues), and escalate-pause (stop and hand over). Escalation is two actions rather than one because notify and pause carry different costs: a needless pause can let a live lead go cold, while a needless notify only spends a person’s attention. Collapsing them would hide that difference from the decision rule.

Cost. A wrong action has a cost, and the costs are deliberately asymmetric. The ordering, from most to least expensive, is: a false assertion, a lead who needed a human but was held, a needless pause, a needless notify, a needless ask. Correct actions cost nothing, except a correct pause, which carries a small residual so the policy does not treat pausing as free and over-use it. The false assertion is not merely the most expensive error; on the cases that forbid a direct answer, such as a request for legal or land documents, the answer action is removed from the choice set entirely. It is a hard constraint, not a high price, because there is no belief state that should make the agent send a false legal claim. The full matrix and the reasoning behind each value are in Section 4.

Policy. The policy computes the expected cost of each available action under the current belief and picks the lowest. It is myopic: it reasons one step ahead and does not plan over how the belief might change on future turns. This is enough for the answer, hold, and escalate actions, whose cost can be judged against the current belief. It is a poor fit for the ask action, whose value is mostly the better belief it buys next turn, which a one-step rule cannot see. This tension is left in the open and discussed in Section 8 rather than papered over.

Feedback. After acting, the agent would in principle observe whether the lead converted or whether an escalated case was resolved. That signal is delayed, noisy, and partly counterfactual, and a missed escalation has no natural label at all: nobody records the lead that needed a human and did not get one. For the experiment in this paper, feedback is supplied by labeled synthetic cases instead, which is what makes recall on missed escalations measurable at all.

Human-reasoning function. The one human-like function the agent adds is to recognise when a decision is too costly to make alone and route it to a person, rather than always acting or always deferring. It maintains an explicit belief, updates it as evidence arrives, and escalates specifically when the expected cost of acting alone is high. This is what makes escalation a costed decision rather than a free safe harbour, and it is the behaviour the rest of the paper measures.

4 Probability model and decision rule

4.1 Belief

The agent never observes a lead’s true intent. It observes one message (optionally with a small amount of conversation context, the turn index and repeat count) and forms a belief over two hidden components.

The first is *readiness*, a distribution over three states:

$$b_r = (P(\text{hot}), P(\text{warm}), P(\text{cold})), \quad \sum_s P(s) = 1. \quad (1)$$

The second is *needs-human*, a single independent probability that the case requires a person regardless of how ready the lead is:

$$b_h = P(\text{needs_human} = \text{True}) \in [0, 1]. \quad (2)$$

These are kept separate on purpose. A lead can be ready to buy *and* need a human (someone has to confirm a booking),

Action	needs_human = False			needs_human = True		
	hot	warm	cold	hot	warm	cold
answer	0	0	0	10	10	10
ask	2	2	2	4	4	4
hold	6	1	0	8	4	3
escalate_notify	3	3	3	0	0	0
escalate_pause	6	5	5	2	1	1

Table 1: Cost matrix $C(a, s, h)$. Costs are expert-set; the ordering encodes that a missed escalation is far more expensive than a needless one. Correct actions cost 0; a correct escalate-pause carries a residual cost of 1 so the policy does not treat correct pausing as free. On `no_direct_answer` cases the *answer* action is removed from the feasible set, so its row does not apply.

or clearly not ready *and* still need a human (an abusive or competitor message a bot should not handle). A single confidence score cannot represent either case. Treating readiness and needs-human as independent is a modelling assumption, not a measured fact, and Section 6 does not validate it: the failures localise to one miscalibrated marginal, which is consistent with independence but not evidence for it, since the run never observes the joint.

Because the two are independent, the joint probability of any full state factors as a product:

$$P(s, h) = P(s) \cdot P(h), \quad (3)$$

where $s \in \{\text{hot}, \text{warm}, \text{cold}\}$ and $h \in \{\text{True}, \text{False}\}$. This gives six joint states. For the worked case in Section 4.3 the belief $b_r = (0.40, 0.50, 0.10)$ and $b_h = 0.30$ expand to $P(\text{hot}, F) = 0.28$, $P(\text{warm}, F) = 0.35$, $P(\text{cold}, F) = 0.07$, $P(\text{hot}, T) = 0.12$, $P(\text{warm}, T) = 0.15$, $P(\text{cold}, T) = 0.03$, which sum to 1.

4.2 Actions and costs

The agent chooses one of five actions: *answer*, *ask* (a qualifying question), *hold*, *escalate-notify* (tell a human while the conversation continues), and *escalate-pause* (stop and hand over). Each action has a cost in each of the six joint states, giving the 5×6 matrix $C(a, s, h)$ in Table 1.

The costs are expert-set, not learned. An earlier draft claimed their *ordering* was the load-bearing part and not the exact magnitudes; a sensitivity sweep supports that for one cell and refutes it for another, so the claim is stated more narrowly here. Sweeping the false-assertion cost from 7 to 30 preserves every ordering, yet the decisions shift midway: 16 misses and 43 escalations from 7 to 10 (cost 1.920 to 1.650), then 14 and 54 from 12 to 30 (1.590 to 1.520). Ordering-invariance holds locally, not across the range. The needless-notify cost is not like that. Sweeping it from 1 to 6, again preserving every ordering, moves the policy across almost its whole operating range: 1 miss and 96 escalations at a cost of 1, 8 and 73 at 2, 16 and 43 at 3, 23 and 28 at 4, and 25 and 23 at 5 or above. The reported operating point is a direct consequence of that one number’s magnitude, and it was set by judgment. Two cells are worth stating because they are where the two-part belief does its work. When a lead is hot *and* needs a human and the agent holds, the cost is 8, not

16: the two harms (the lead waits, no human arrives) overlap rather than stack, and pricing it at $6 + 10$ would exceed the false-assertion cost and break the framing in which a false assertion is the top of the scale. When that same lead is answered, the cost is 10 with no premium for hotness, since a false claim is equally false to a hot or a cold lead.

The false assertion is not merely priced high; it is a *hard constraint*. On the cases that carry a `no_direct_answer` flag, the *answer* action is removed from the feasible set rather than charged a high price. A price can be outweighed if enough other costs pile up; a constraint never can. There is no belief state that should ever make the agent send legal or land documents, so this is enforced as infeasibility, not as a large number.

4.3 Decision rule

The policy is *myopic and one-step*: it picks the action with the lowest expected cost under the current belief, with no lookahead over future turns. For a feasible action a ,

$$\mathbb{E}[C(a)] = \sum_s \sum_h P(s) P(h) C(a, s, h), \quad (4)$$

and the chosen action is

$$a^* = \arg \min_{a \in \mathcal{A}_{\text{feasible}}} \mathbb{E}[C(a)]. \quad (5)$$

For the worked case above this yields expected costs of escalate-notify 2.10, ask 2.60, answer 3.00, hold 3.68, escalate-pause 4.20, so the agent escalates, and the realised cost is 0 because the case truly needed a human.

4.4 Where the threshold comes from

Because the readiness rows for *answer* and *escalate-notify* are flat across readiness (a false answer costs 10 in any readiness state where a human is needed; a needless notify costs 3 in any state where one is not), the choice between them reduces to a single threshold on b_h . Answer is cheaper while $10 b_h < 3(1 - b_h)$, and escalate-notify takes over when the inequality flips, at

$$b_h^* = \frac{3}{13} \approx 0.2308. \quad (6)$$

This crossover is independent of the readiness distribution, precisely because both rows are flat across readiness. In the worked case $b_h = 0.30$ clears it by 0.069, which is why the agent escalates even though the belief is only modestly above the line.

The precision of the fraction is misleading, however, and it is worth saying so before the results lean on it. The elicited b_h takes only eight distinct values over the 100 cases, all at one decimal place, and none of them falls in $(0.2, 0.3]$. Every threshold in that interval therefore produces identical decisions on every case, and $3/13 \approx 0.231$ is one arbitrary point inside a band the measurement cannot resolve. Nothing in the results distinguishes it from 0.25.

This clean single threshold holds only when the runner-up action is *answer*. When *hold* is the competing action its row is not flat across readiness, so the threshold becomes readiness-dependent, and the spread is much wider than the two hold cases in the failure analysis (which flip at 0.306 and 0.373)

would suggest. Solving $E[\text{notify}] = E[\text{hold}]$ at each case’s actual readiness distribution puts the crossover anywhere in $[0.018, 0.500]$ on the 73 cases where it falls in $[0, 1]$ at all; on the other 27 it is negative, meaning *hold* never wins for any admissible b_h . At the pure readiness states the crossover is -0.600 (hot), 0.333 (warm) and 0.500 (cold). So $3/13$ is the threshold for one specific pairwise comparison, not a single operating point for the policy. The policy is still a single $\arg \min$; it is only the *reduced* one-dimensional threshold that shifts with which action is the alternative, and quoting $3/13$ alone understates how much of the decision boundary depends on readiness.

4.5 Baseline

The baseline is the same decision rule run over an identical belief with a *uniform* cost matrix, every non-zero cost replaced by one. This isolates exactly what cost-awareness buys: the belief, the feasible-set constraint, and the expected-cost machinery are all held fixed, and only the asymmetry is removed.

5 Test method

5.1 Cases

The agent is tested on 100 synthetic cases. Each case is one inbound message with a label for the true readiness (hot, warm, cold) and whether it truly needs a human, plus the position context (turn index and repeat count) the belief sees. The cases are built from eleven archetypes that between them cover the situations where the cost trade-off actually bites: template openers, requests for photos, one-word messages, ready-to-book buyers, requests for legal or land documents, messages that suspect the agent is a bot, competitors fishing for pricing, captionless media, over-sharers, polite time-wasters, and abusive or off-topic messages. The distribution is deliberately weighted toward the archetypes where a wrong action is expensive rather than made uniform, and it is skewed toward harder cases: 42 of the 100 truly need a human, chosen so that recall on missed escalations is measurable at all. How far that is above a live inbox is not known here — no measured base rate for this channel is available to the author — so the 42% should be read as a design choice made for measurability, not as a calibrated departure from a known population.

The readiness distribution of the case set is also worth stating explicitly, because it does not match the prior the belief module itself carries. That prior is (hot, warm, cold) = $(0.02, 0.13, 0.85)$, whereas the true labels in the case set are 23 hot, 38 warm and 39 cold. The set is not a sample from the population the design assumes; Section 6 reports what happens when the results are reweighted toward that prior.

The labels are the author’s judgment of the intended state, not outcomes observed in production. This is stated plainly as a limitation in Section 8: the readiness and needs-human labels are what the message was written to represent, not what a real lead later did.

5.2 Beliefs

For each case the belief is produced once by a language model (gpt-4o-mini) and cached to disk, keyed by case id. Both

policies are then run over the identical cached beliefs, so any difference between them comes from the decision rule and not from variation in the belief. Caching also makes the run reproducible: the language model is non-deterministic, so regenerating the beliefs would move every reported number, and the cache is committed alongside the code for that reason.

5.3 Split

The 100 cases are split 50/50 into a development and a test half with a fixed seed, stratified by archetype and sub-variant so that every variant appears in both halves and the needs-human count is balanced (21 in each). The split is not redrawn per run. Because the halves are matched by construction, close agreement between development and test totals reflects that construction rather than out-of-sample generalization, and Section 6 reads the split numbers with that in mind.

5.4 Policies and baselines

Two policies are compared over the same beliefs. The *cost-aware* policy is the minimum-expected-cost rule of Section 4 under the full asymmetric cost matrix. The *uniform* baseline is the same rule with every non-zero cost replaced by one, which holds the belief, the feasible-set constraint, and the expected-cost machinery fixed and removes only the asymmetry. This isolates what cost-awareness buys. Three fixed-action baselines are also reported for reference: always answer, always ask, and always escalate-notify.

One implementation detail changes a reported number, so it is disclosed here rather than left to the failure analysis. Exact ties in expected cost were originally resolved in the order the five actions are declared, which resolves toward *answer* — the action with the largest worst-case cost, and the one a policy premised on the expense of false assertion should reach for last. Ties are now resolved safest-first, in an order derived from the cost matrix itself by worst-case cost (*tie_break_order*) rather than hand-written. The run reported below deliberately retains the original order, via the `--legacy-tie-break` flag, so the committed result artifact stays reproducible action for action. Under the corrected default exactly one of the 100 decisions changes and the mean cost falls from 1.72 to 1.65; Section 7 gives the case and the arithmetic.

5.5 Metrics

Accuracy alone would hide the thing the design is about, so the run reports mean realised cost per case, the number of missed escalations (a case that needed a human but did not get one), escalation precision and recall, the number of escalations, the number of hard-constraint violations, and the calibration of the belief. Calibration is measured as expected calibration error on the needs-human marginal, since that is the probability the escalation threshold acts on.

6 Results

Table 2 reports all five policies over the 100 cases, on the shared cached beliefs.

Policy	Cost	Miss	Prec.	Rec.	Esc.
cost-aware	1.72	16	0.605	0.619	43
uniform baseline	2.58	24	0.750	0.429	24
always-notify	1.74	0	0.420	1.000	100
always-ask	2.84	42	—	0.000	0
always-answer	3.40	34	1.000	0.190	8

Table 2: Mean cost per case, missed escalations, escalation precision and recall, and number of escalations, over 100 cases. Lower cost and fewer misses are better. The cost-aware row is the legacy tie-break configuration described in Section 5; under the corrected default its cost is 1.65 and every other entry in the row is unchanged. No policy records a hard-constraint violation, including the fixed-action baselines — the constraint forbids only *answer*, and on all 8 restricted cases the cost-aware policy escalates on cost grounds before feasibility is consulted, so the constraint never changed a decision. Zero violations here is therefore not evidence that the constraint mechanism works; it is untested. *always-answer* shows precision 1.000 on 8 escalations, which is an artifact: the policy never escalates, and those 8 are the restricted cases where the constraint removed *answer*, leaving the fixed-action policy to fall back to escalate-notify by construction rather than by any comparison of costs. All 8 happen to need a human, so the precision is 8/8 on a denominator the policy did not choose. It should not be read as the baseline discriminating well.

6.1 What cost-awareness buys

The cost-aware policy reaches a mean cost of 1.72 against 2.58 for the uniform baseline, a reduction of a third over 100 shared beliefs. Because the two policies differ only in whether the cost matrix is asymmetric, this gap is what cost-awareness buys and nothing else. The two disagree on 46 of the 100 cases, and the cost-aware policy is not cheaper on more of them: it is cheaper on 15, the baseline is cheaper on 18, and 13 are ties. It wins overall because its losses are small and its wins are large. Almost every swap it makes is answer to hold or answer to escalate-notify, trading many small costs of two or three for a few avoided false assertions of ten. The point of the design is not to be right more often but to be wrong more cheaply.

6.2 What the baseline actually is

One further check weakens how much that gap can be attributed to the machinery. The uniform baseline selects the same action as a plain threshold of 0.5 on b_h — ignoring readiness, the cost matrix, and the expected-cost computation entirely — on 100 of the 100 cases. Its behaviour is fully described by one number. The cost-aware policy is not reducible in the same way, because *hold* is chosen on 27 cases and that choice does depend on readiness, but the comparison in Table 2 is between a readiness-sensitive rule and a 0.5 threshold, and part of the reported gap is the distance from 0.5 to 3/13 rather than the asymmetry as such.

A threshold sweep separates the two contributions, and it does not go cleanly. No threshold on b_h reproduces the cost-aware policy’s 43 escalations: because the elicited values are quantized, the sweep jumps from 29 escalations at $t = 0.35$ (23 misses, cost 2.600) to 46 at $t = 0.25$ (13 misses, cost 1.810). The cost-aware policy’s 16 misses at 43 escalations is

therefore not straightforwardly better than a tuned threshold at matched load, because matched load is not attainable; the nearest bracketing thresholds are worse on one axis and better on the other. Tuning a threshold on the development half by realised cost selects $t = 0.00$ — escalate everything — at cost 1.740, which recovers the always-notify result rather than the cost-aware one.

6.3 Two actions that are never used

Of the five actions, the cost-aware policy selects only three across all 100 cases: *answer* 30 times, *hold* 27, *escalate-notify* 43. *Ask* and *escalate-pause* are selected zero times. This is a property of the matrix rather than of the cases. *Ask* costs 2 when no human is needed against 0 for *answer*, and 4 when one is needed against 0 for *escalate-notify*, so it is dominated at both ends and can only win in a narrow middle band that the quantized beliefs never land in. *Escalate-pause* is dominated by *escalate-notify* in every column. The paper describes a five-action set, but every reported number comes from a three-action policy, and the two unused actions are untested by this experiment. The myopia limitation in Section 8 compounds this: a one-step rule cannot price *ask*’s payoff, so the action most obviously suited to an ambiguous inbound message is both unpriced and unused.

6.4 The comparison that is a tie, not a win

Against always-notify, which escalates every message, the cost result is close to a tie: 1.72 against 1.74. This is not a mistake in the design, it is a property of the cost matrix. A needless notify costs 3 and a false assertion costs 10, so escalating everything is cheap and is barely punished; blanket escalation almost never pays the expensive error. The honest claim here is therefore not that the cost-aware policy is cheaper than always-notify, because on cost it essentially is not. What it buys is human load. It reaches the same cost while escalating 43 times out of 100 rather than all 100, at a precision of 0.605 against 0.420. The same outcome without dragging a person into more than half again as many conversations is the operational win, and it is a different claim from a cost win, so both are reported.

6.5 The development and test split

Both halves reach the same total cost for the cost-aware policy (86 each) and for the uniform baseline (129 each). This looks like clean generalization and is not. The split was stratified by archetype and sub-variant and balanced on the needs-human count, so the two halves are matched by construction and the totals coincide for that reason. The policies do make different decisions on the two halves (the escalation precision differs across them), so the halves are not identical case-for-case; the aggregate totals simply agree because the halves were built to mirror each other. On synthetic data of this kind, matched by design, this figure cannot stand as out-of-sample evidence, and it is not read as such.

6.6 Calibration

Every one of the 16 missed escalations came from the same place: the belief under-estimated the needs-human probability, not the readiness. On all 16, the needs-human estimate

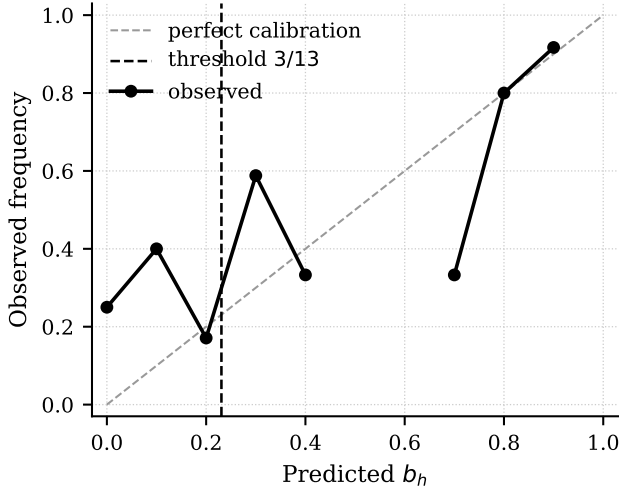


Figure 1: Reliability of the `needs_human` marginal over all 100 cases, ECE 0.142 (bootstrap 95% CI [0.100, 0.249], median 0.171). Points above the diagonal are under-confident: the belief reports a low probability where the observed frequency is high. Marker area is proportional to bin count and bars are Wilson 95% intervals, because the bins hold between 4 and 35 cases and most are individually consistent with perfect calibration. The vertical line is $b_h^* = 3/13 \approx 0.231$; the shaded band is the interval of thresholds that decide identically, since the elicited marginal takes only one-decimal values and no case falls in $(0.2, 0.3]$. The bin that *contains* the threshold, 0.2–0.3, is the largest ($n = 35$) and is very slightly over-confident (predicted 0.200, observed 0.171, CI [0.081, 0.327]). Its neighbours at 0.1–0.2 and 0.3–0.4 are under-confident by 0.300 and 0.288. The 0.7–0.8 bin reads over-confident by 0.367 on 6 cases, but its interval [0.097, 0.700] covers the predicted value, so the belief is not uniformly under-confident and that bin is not evidence of the reverse. Bins 0.5–0.7 are empty, hence the break in the curve.

was 0.30 or below against a true label of needing a human, while the readiness `argmax` on those same cases was spread across cold, hot, and warm and was often correct and confident. The belief reads under-confident on the needs-human marginal, with an expected calibration error of 0.142 (bootstrap 95% CI [0.100, 0.249]; 0.168 on the development half, 0.184 on test), and reliability bins (Figure 1) that read low against observed frequency near the escalation threshold (for example, a predicted 0.10 against an observed 0.40, and a predicted 0.30 against an observed 0.588).

Two qualifications belong with that number. First, the bin that actually *contains* 3/13 is not the under-confident one: at 0.2–0.3, the largest bin at $n = 35$, predicted 0.200 against observed 0.171, the belief is very slightly *over*-confident, and its Wilson interval [0.081, 0.327] covers the predicted value. The under-confidence is in the neighbouring bins. So “under-confident exactly where the threshold sits” is not the right description; the mechanism is that a large mass of cases sits pinned at $b_h = 0.20$, just below the line, and 35% of the case set turns on whether that single elicited value is right. Second, with 4 to 35 cases per bin most individual bins are consistent with perfect calibration, so the ECE point estimate carries most of the evidential weight here and its interval is

wide.

Because the failures trace to the needs-human marginal rather than to the readiness marginal, recalibrating that marginal is the natural first fix, and Section 7 reports what it recovers. This is weaker than the claim an earlier draft made. That the misses localise to one marginal is consistent with the independent two-part belief, but it is not evidence *for* independence: the experiment records only the two marginals, never the joint, so it cannot detect the two components interfering even if they did. Testing independence would require eliciting $P(s, h)$ directly and comparing it against $P(s)P(h)$, which this run does not do.

7 Failure analysis

The cost-aware policy makes 16 missed escalations and no other class of error worth naming: no hard-constraint violations, and its needless actions are the cheap ones the design is willing to pay. So the failure analysis is almost entirely about the misses. Five are examined in detail below, chosen to cover the distinct ways a miss happens.

7.1 The shared cause

All 16 misses have the single cause described in Section 6: an under-estimated needs-human probability. None of them come from the two hidden components being confused with each other. This matters for how the misses should be counted. Treating every miss as a false assertion at cost 10 overstates the damage, because a *hold* is a partial hedge: it is the wrong action, but it does not deliver a false answer, so the matrix charges it 3 or 4 rather than 10. Across the 16 misses the realised costs are nine at 10, four at 3, and three at 4, for 114 in total, not the 160 that costing every miss at 10 would give. One caveat belongs with this diagnosis. That every miss traces to an under-estimated needs-human probability is a statement about this labelling scheme, not a proof of the underlying causal mechanism: the experiment does not independently exclude cost-model misspecification, the coarse three-state readiness space, or the one-decimal quantization of the belief as contributing causes, and the recalibration in §7.3 is an in-sample ceiling rather than a held-out test.

7.2 Five cases

Table 3 lists five misses and the mode each one represents.

The three that realise the full cost of 10 fail in different ways. `a04-booking-042` is the cleanest illustration of the shared cause: the readiness estimate was right and confident, and only the needs-human number failed, so nothing about readiness could have saved it. `a02-deep-015` sits one quantization step below the line: its elicited b_h is 0.20 against a threshold of 0.2308. Describing this as missing by 0.031, as an earlier draft did, reads the gap as a small numerical shortfall, which overstates how close the call was. The belief cannot express 0.24; the next value it can emit is 0.30. Recovering this case requires moving it a full decimal step, not nudging it by three hundredths, and the same is true of the 35 cases that share its $b_h = 0.20$. `all-repeated-097` is the worst of the three, for a reason that is not the cost number — and, unlike the other four cases in Table 3, it is not an

Case	Cost	Mode
a11-repeated-097	10	tie-break, not the cost model; since corrected
a04-booking-042	10	readiness right, needs-human wrong
a02-deep-015	10	one quantization step below the line
a03-followup-024	4	hold looked cheap, readiness also off
a10-persistent-091	3	cheapest miss the matrix allows

Table 3: Five missed escalations, one per failure mode. All five are from the reported run. The first has since been fixed in the implementation and is retained because the reported run predates the fix; the other four remain open.

open failure: its cause was identified and repaired, and it is retained here because the reported run predates the repair. Its belief is the furthest from the flip point of any miss: it needed a needs-human estimate above 0.50 and the model reported 0.00, so recalibration would not save it the way it would save the near-threshold cases. And it was not decided by the cost model at all. Two actions came out with an expected cost of exactly 0.000, and the tie was broken by the fixed ordering of the action list rather than by any cost comparison. In the true state, holding would have cost 3, so 7 of its 10 realised points were attributable to an implementation detail, the tie-break, rather than to a property of the policy.

That detail has since been repaired, as Section 5 notes. Ties now resolve safest-first, in an order derived from the cost matrix by worst-case cost (`tie_break_order`) rather than hand-written, which gives *escalate-notify* (worst case 3) < *ask* (4) < *escalate-pause* (6) < *hold* (8) < *answer* (10). The old ordering resolved indifference toward *answer*, the action with the largest downside, which is the opposite of what a policy premised on the expense of false assertion should do. Exactly one decision on the reported case set changes: *a11-repeated-097* moves from *answer* to *hold*, and the mean cost falls from 1.720 to 1.650 — the 7 points this section predicted, and no others. Miss count and escalation count are unchanged at 16 and 43, since *hold* and *answer* are both non-escalating actions. This is also the only decision the correction *could* change: *a11-repeated-097* is the one case in the 100 where two feasible actions come out at exactly equal expected cost, and a tie-break is consulted nowhere else. The uniform baseline is unaffected, for a structural reason worth stating — with every non-zero cost set to one, all five actions have worst-case cost 1, so the derived order degenerates to the declaration order and the baseline’s 2.58 is identical under both rules. Table 2 reports the pre-fix numbers, because `results/run.json` was generated before the change and is kept as the artifact the reported numbers are checked against; `run_policies.py --legacy-tie-break` reproduces it exactly, action for action on all 100 cases, so those numbers stay verifiable while the default path uses the corrected rule.

The two cheaper misses show why not every miss

is a false assertion. *a03-followup-024* and *a10-persistent-091* were held rather than answered, so they cost 4 and 3 rather than 10. For these the binding comparison is hold against *escalate-notify*, not answer against *escalate-notify*, and because the hold row is not flat across readiness their thresholds are readiness-dependent rather than the clean 3/13 of the answer cases. Reporting them at their true cost, rather than rounding every miss up to 10, is what keeps the miss total honest.

7.3 What recalibration actually recovers

Since the diagnosis above points at the needs-human marginal, it can be tested directly rather than left as a prediction. Replacing each elicited b_h with the observed frequency of its reliability bin — histogram binning, fit on the same 100 cases, so this is an optimistic in-sample ceiling and not a held-out result — and re-running the policy gives mean cost 1.250 against the corrected baseline of 1.650, and 8 misses against 16.

The composition of that change matters more than the totals. Of the 16 original misses, 9 are fixed and 7 survive. The survivors are the cases the diagnosis already flagged as out of reach: 6 sit at $b_h = 0.20$, in the bin whose observed frequency (0.171) is itself below the threshold, so recalibration moves them the wrong way, and 1 is *a11-repeated-097* at $b_h = 0.00$. Recalibration also *creates* one new miss: *a06-frustrated-055*, at $b_h = 0.70$, is pulled down into the over-confident 0.7–0.8 bin and switches from *escalate-notify* to *hold*.

The cost is paid in human load. Escalations rise from 43 to 60, precision falls from 0.605 to 0.567, and recall rises from 0.619 to 0.810. Recalibration does not make the policy better on every axis; it buys recall with load, and 8 misses remain that no amount of recalibrating this marginal removes. That floor, not the halving, is the finding.

7.4 A failure the test set cannot show

One failure did not come from the synthetic run at all. In a live deployment a lead sent a routine request for information and, before the agent had replied, a second short message asking to be contacted directly by a person. The agent acted on the routine message and silently dropped the second, which carried the only needs-human signal in the turn. The rare, high-cost signal was averaged away by the common, low-cost one because the two messages were handled as a single turn.

This is a different kind of failure from the 16 misses. Those are belief-scoring errors on a single message. This one is a turn-boundary error: the important message never received its own belief evaluation. The synthetic test set cannot produce it by construction, because every case carries one message as a single string and the harness scores one message per turn. Surfacing this class of failure would need multi-message-per-turn cases and a per-message evaluation step before the turn is acted on, neither of which this experiment has, so none of the numbers reported here bear on it. It is recorded here, and as a limitation in Section 8, rather than left out because the harness happens not to reach it. It is also the failure that several practitioners, on being shown it, independently proposed

the same fix for: scan each message for a small set of critical intents before the turn is scored as a whole, so that a rare handoff signal overrides rather than averages into the turn.

8 Limitations, ethics, and human control

8.1 Limitations

The results rest on synthetic data, and several of the paper’s numbers are only measurable because of that. The clearest case is recall on missed escalations. In a live inbox there is no label for the lead who needed a human and did not get one, so escalation recall cannot be measured at all; it is measurable here only because the cases are labeled by construction. The needs-human base rate of 42% was chosen so that recall is estimable. Earlier drafts described it as inflated above a real inbox; that comparison is withdrawn, because no measured base rate for this channel exists to compare against. What can be said without a source is narrower and still sufficient: the rate was set for measurability rather than sampled, so the precision and recall reported here are properties of this case set and should not be expected to transfer to production, in either direction. The readiness distribution is off the design’s own prior in the same way: the belief module assumes 85% cold, the case set is 39% cold. The readiness and needs-human labels are the author’s judgment of what each message was written to represent, not outcomes observed from real leads, so they carry the author’s assumptions about intent.

The state space is deliberately coarse and this costs the model in two visible ways. Non-leads such as competitors and abusive messages are labelled cold rather than given their own state, which slightly distorts readiness calibration. And the state cannot separate a competitor fishing for pricing from an abusive message: both come out as cold and needing a human, yet the two call for different actions. The state space simply cannot see the difference, which is a real finding about its granularity rather than a bug in the policy.

Two limitations are structural to the method rather than to the data. The policy is myopic, so it undervalues the ask action, whose payoff is a better belief on the next turn than a one-step rule cannot price; the honest handling of the ask action would compute its one-step value of information, which this design does not do. Section 6 shows the consequence is not hypothetical: *ask* is selected zero times, as is *escalate-pause*, so the reported results exercise three of the five actions. And the hard constraint that removes the answer action on restricted cases is treated here as observable and error-free, because a flag on the case says the message asks for legal or land documents. A production detector for that condition would carry its own false-positive and false-negative rates, which this experiment does not model. Sweeping a synthetic false-negative rate on that flag from 0 to 0.5 changes nothing — cost stays at 1.650 and misses at 16 — but that is not reassurance, it is a restatement of the fact that the constraint never bound: on all 8 restricted cases the policy already escalates on cost grounds, so dropping the flag changes no decision. A detector’s error rates would matter on restricted cases the cost model does *not* already catch, and this case set contains none.

A third structural limitation is the granularity of the belief itself. The elicited b_h takes only 8 distinct values across the

100 cases, every one at a single decimal place, distributed {0.0: 4, 0.1: 15, 0.2: 35, 0.3: 17, 0.4: 6, 0.7: 6, 0.8: 5, 0.9: 12}. The language model is being asked for a probability and is returning something closer to an ordinal grade. Three consequences follow, and all three are visible in the results: the threshold 3/13 cannot be distinguished from any other value in (0.2, 0.3]; 35% of the case set rests on one elicited value, $b_h = 0.20$, sitting just below the line; and recalibration cannot rescue those 35 cases, because it maps a bin to a single number and that number (0.171) is also below the line. A finer-grained elicitation is the prerequisite for most of the improvements this paper proposes, and it is not tested here. Relatedly, the beliefs come from a single unpinned model (gpt-4o-mini) elicited once per case; no dated snapshot is recorded, no repeat elicitation measures the variance of the belief itself, and every number in the paper is conditional on that one draw.

Because the case set’s readiness distribution does not match the prior the design assumes, the reported means are sensitive to reweighting. Importance-weighting each case toward the module’s own prior (weights: hot 0.087, warm 0.342, cold 2.180) moves the cost-aware policy from 1.720 to 1.456 and the uniform baseline from 2.580 to 2.526 — the gap widens from 0.86 to 1.07. It also reverses the comparison against always-notify, which moves from 1.740 to 1.834. The direction of the main result survives reweighting; its magnitude does not. Since the prior itself is expert-set rather than measured, neither weighting is the authoritative one, and the spread between them is the honest statement of what the case set can support.

Finally, one whole class of failure is invisible to the test set by construction. As Section 7 describes, a high-cost signal batched with a routine message at a turn boundary cannot occur in a set where every case is a single message scored on its own. This is the sharpest version of the general point that synthetic data trades realism for measurability: here the trade hides not a distorted number but an entire failure mode.

8.2 Ethics and human control

The system decides when a person is pulled into a conversation, and the costs are set to make missing a needed human more expensive than a needless handoff. That is deliberate, but it means the agent will sometimes escalate cases that did not need it, spending human attention as the price of not missing the ones that did. The hard constraint on false legal and land claims is the other side of this: some answers are treated as a line the agent may not cross at any belief, rather than a cost to be traded off, because the harm of a confident false claim on a consequential matter is not the kind of thing a cheaper action elsewhere should be able to outweigh. The experiment only measures the cost of choosing to escalate. It doesn’t account for what happens afterward—such as human mistakes, limited capacity, or unclear accountability. Those are left for future research. A deeper point raised in the live discussion is worth recording as a direction the current design does not yet meet. The turn-boundary failure is not only a scanning problem; the underlying issue is that a message was droppable at all. A more robust design would make every inbound item durable, with an explicit acknowledge or

close step, so that a request for a human keeps returning until something resolves it, regardless of how a later ranking scores its priority. Priority can be fuzzy; existence should not be. The present design scans for critical signals before the turn is acted on, which reduces the risk, but it does not yet guarantee that no message can be silently dropped, and that guarantee is the right target for a production system.

9 Conclusion and new questions

This paper built a small agent that decides, on each inbound message, whether to answer, ask, hold, or escalate to a human in one of two ways, when the lead’s readiness and whether they need a person are both hidden. The decision rule is a textbook one: minimise expected cost under a belief. The contribution is not the rule but an honest account of what it buys. Making the cost matrix asymmetric cuts mean cost by a third against the same rule without that asymmetry. Against a policy that simply escalates everything, it does not cut cost at all, and the paper says so; what it buys there is reaching the same cost while escalating fewer than half as often. Every missed escalation traces to the needs-human marginal rather than to the readiness marginal, which points at recalibration rather than redesign — though, as Section 6 notes, that localisation is consistent with the two-part belief without being evidence for it, since the experiment never observes the joint distribution.

Three questions follow directly from these results.

The first is calibration, and it is the one this version can partly answer. Recalibrating the needs-human marginal in-sample halves the misses, from 16 to 8, and cuts mean cost from 1.650 to 1.250 — but the composition is 9 fixed, 7 surviving, and 1 newly created, and escalations rise from 43 to 60 to pay for it. The floor is real and it has a specific cause: 6 of the 7 survivors sit at $b_h = 0.20$, in a bin whose observed frequency is also below the threshold, so no mapping of that bin to a single value can rescue them. The floor is therefore set by the granularity of the elicitation, not by the calibration method. The next step is a finer-grained belief and a held-out recalibration fit, since the numbers here are an in-sample ceiling.

The second is the ask action. A myopic policy cannot price the value of asking, because that value is a better belief next turn. This experiment makes the cost of that concrete: *ask* is dominated by *answer* when no human is needed and by *escalate-notify* when one is, so it is selected zero times out of 100, as is *escalate-pause*. The five-action set is really a three-action set in practice. Scoring the ask action by its one-step value of information, and testing whether it then earns its place against hold and escalate, would close the gap the current design leaves open, and would say whether the action is genuinely useful or merely unpriced.

The third is the turn boundary. The one failure the synthetic test set cannot produce is the one that came from a live run, where a high-cost signal was lost when batched with a routine message. Building cases with more than one message per turn, and a per-message scan that lets a rare critical signal override the turn, would let this failure be measured rather than only reported, and would test the durability idea

that no inbound message should be silently droppable in the first place.

AI-use statement

This project was built with AI assistance used as a tool under the author’s direction, not as an author. The author selected the problem, made and locked every design decision, set the cost values and their ordering, wrote all public discussion contributions personally, and verified every reported number against the committed run artifacts.

AI tools were used for: preparing research terms, search queries, and candidate communities and accounts to engage [tools: Claude, ChatGPT, Gemini]; studying the source papers [tool: NotebookLM]; scaffolding, writing, and repairing code, running the test harness, and executing commits [Claude and Claude Code]; and adversarial review passes of the draft [tools used, logged in `review-record.md`]. Per-decision provenance — what the author set versus what a tool proposed — is recorded at the point each decision was made in `build-log.md`.

Every AI output was checked before use. Where a tool was confidently wrong, the error and its correction are recorded rather than propagated; these are logged in the “Important AI errors” section of `research-file.md`. Two are worth naming: a synthesis that recommended an architecture directly contradicting the locked design, and an unverified bound attributed to a mis-cited source, neither of which was carried into this paper.

The author affirms that no conversation in the discussion record is fabricated; that every reported result comes from a real run of the committed code over the committed data; and that no reference is cited that the author did not read.

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